

Integrated Requirements and Systems Modeling in the Mechatronic Development Process

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Abstract—Manufacturing systems continuously increase in complexity due to higher customer expectations regarding their purpose and flexibility of use. In addition, the growing meaning of electronic and software components in manufacturing systems complicates the engineering since the interaction of different disciplines leads to increasing coordination efforts and problems regarding the interfaces between these disciplines. For these reasons, development processes in this context need to be adapted continuously to the present situation. To face these challenges, the suitable handling of arising requirements is one key factor for the success of engineering processes. Therefore, a modeling technique for an integrated requirements and systems modeling in the mechatronic development process is presented in this paper. After a summary of approaches for model-based development and requirements engineering in the context of mechatronic systems, the modeling technique is illustrated on a meta-model level. To demonstrate the feasibility, the proposed approach is applied to a miniaturized packaging machine, which is currently developed at the Institute for Machine Tools and Industrial Management (*iwb*) of Technische Universität München for benchmarking engineering methods and tools. Excerpts of this application are outlined in this paper as well as the results of a discussion with experts from the machine tools industry regarding the presented modeling approach. Finally, an outlook with respect to possible enhancements of the modeling technique is given.

I. INTRODUCTION

The engineering of machines and plants is currently subjected to global mega-trends like growing customer-individuality in products or shortening of innovation cycles [1]. Besides technological improvements, development processes form a key factor to meet current and future challenges (e.g. increasing interdisciplinarity) [2]. Since software is already the main innovation driver in many industry branches [3], the still mechanically dominated engineering principles have to be substituted by mechatronic development approaches, which allow an interdisciplinary engineering [4]. Especially the interaction of the main engineering disciplines (i.e. mechanics, electrics and software) leads to various problems, which can be reflected in a poor product quality or delays in the commissioning [5].

To overcome this situation, the model-based development of mechatronic systems is one common approach [4]. In contrast to the application of single engineering models (e.g. from CAD-tools), model-based mechatronic development means the

alignment of the whole engineering process with an interdisciplinary model of the system [6]. Through the elaboration of such models, developers from the main engineering disciplines gain a common understanding of the mechatronic system and are able to contribute their competencies and expertise [4]. In addition, the application of a model-based development approach allows the integration of further development tasks, like for example requirements engineering. Especially in a machine or plant development process, where requirements are usually handled with textual documents, this integration has a major potential to enhance the requirements engineering [7]. This is necessary, since the fulfillment of customer requirements is a decisive factor to stay successful on the market [8].

Therefore, this paper presents a modeling technique that allows an integrated depiction of requirements and systems during early phases of the mechatronic development process. The paper is structured as follows: In Section II the state of the art is briefly reviewed. On this basis, the modeling technique is outlined on a meta-model level, wherein especially the individual aspects for the depiction of requirements are focused (Section III). Subsequently, Section IV illustrates the application and evaluation of the modeling technique using an industrial-close case study. Finally, the paper finishes with a brief summary and an outlook to further activities like the extension of the modeling technique or the evaluation of the entire approach using the example of a machine tool.

II. STATE OF THE ART

In the following subsections current approaches for model-based development and requirements engineering regarding mechatronic systems are summarized in order to indicate deficiencies and derive potentials for advances.

A. Model-Based Development for Mechatronic Systems

Model-based development principles originated in the field of software engineering. Especially object-oriented modeling tools and the related Unified Modeling Language [9] contributed to a broad acceptance [10]. In general, the objective of model-based software development is to supply the engineering process with the targeted use of models, which allow an abstract and easy-to-understand description of software and an automatic generation of the respective source code [10].

Whilst model-based development principles already are well established in the domain of software engineering, their application within the field of mechatronic systems is still in an initial stage [11]. However, the main objective is similar to that in software engineering: By using interdisciplinary models of a mechatronic system, the collaboration of all involved engineers is simplified since a common understanding of the system to be developed is generated [6]. For this purpose, various approaches have been developed in the scientific environment, which are briefly presented in the following.

A first step towards an interdisciplinary mechatronic development is a consistent and abstract description of the system to be developed. Therefore, numerous modeling languages like the SysML [12] have been developed up to today. Particularly, function-oriented methods including the example from Kallmeyer [13], which deal with the depiction of system functions, are getting more attention as they permit a solution-neutral modeling. Such modeling languages form the basis for an integrated model-based development. Eigner et al. [4], for example, extend the well-known V-Model from [14] in all development phases with model-based development methods using the mentioned language SysML. Especially the systems specification and analysis are focused to allow early virtual tests of the respective product. A comparable approach was focused within the research projects Föederal [15] and AQUIMO [16]. While the first research project dealt with the set-up of an information architecture for mechatronic system development, in AQUIMO an engineering tool was developed, which allows a formal modeling of interdisciplinary information. In addition, a process model was outlined, which describes how the engineering tool is used throughout a development process. Another approach for an easy modeling of mechatronic systems was illustrated by Gausemeier et al. [17]. The specification technique CONSENS allows an interdisciplinary description of mechatronic systems using eight different views on a mechatronic system also including a view for requirements and use-cases. In addition, there exist numerous approaches that also contribute to a model-based development of mechatronic systems. Examples include the approach from Thramboulidis [18], who introduced different abstraction layers for mechatronic systems, or from Lüder et al. [19], who outlined a mechatronic modularization of manufacturing systems in order to enable an easy modeling and system analysis. Finally, several methodologies were developed in the field of model-based systems engineering. Estefan [6] summarizes important approaches including the INCOSE object-oriented systems engineering method [12].

B. Requirements Engineering for Mechatronic Systems

Like model-based development, also requirements engineering has its origin in software engineering [20]. In this field, various approaches were developed during the last years in order to enable a deliberate handling of requirements. Davis [21], for example, defines four important subsets of requirements engineering activities: requirements elicitation, triage, specification and change management. In addition, he suggests different methods and techniques for each subset (e.g. user interviews for the elicitation) [21]. Pohl & Rupp [22] also propose various methods for the requirements elicitation, but focus especially on the requirements documentation distinguishing between natural-language and model-based documentation.

Besides these approaches also individual procedure models have been developed including the model from Hood & Wiebel [23], who define an iterative cycle of elicitation, specification, analysis and review for the definition of requirements.

In the approaches presented above the requirements engineering is defined as a stand-alone development activity. However, the multi-disciplinary nature of mechatronic systems necessitates a requirements engineering integrated in the systems development process [20]. Regarding this field, especially the interface between requirements and systems engineering was considered in detail, like for example by Hood et al. [24]. But even mechatronic systems have been focused including the approach from Schedl [20]. She defines a three-phase procedure consisting of an elicitation planning phase, a description phase and a requirements usage phase throughout the development process. In addition, Jung [25] focuses especially on the requirements analysis in interdisciplinary development environments using an iterative development method. Further examples include the approach from Reiss & Schuller [26] proposing a workflow-based approach for requirements engineering or from Gürtler et al. [27], who integrate the requirements analysis in the system conception phase using the example of a product-service-system.

C. Remaining Challenges

The preceding sections show that various efforts have been made to improve the development of mechatronic systems. Model-based approaches have great potential to encounter the difficult collaboration and synchronization of engineering disciplines in particular in early engineering stages [4]. However, these approaches do not focus appropriately on the requirements engineering as an essential part of mechatronic development processes. In consequence, the elicitation, handling and evaluation of requirements has to be carried out as a detached development activity neglecting possible synergy effects [20].

On the other hand, some of the approaches presented in the last subsection address these synergy effects by combining requirements and systems modeling within mechatronic development processes. These approaches allow an integrated depiction of requirements and systems and thus guarantee an appropriate consideration of requirements throughout the entire development process. However, this integration is mostly limited to generic development methods and procedure models. Especially a model-based implementation of development processes is not considered. Thus, further advantages, like an automated requirements verification using system simulation, cannot be applied.

As a conclusion it can be said that both model-based development and requirements engineering can contribute to a purposeful development of mechatronic systems. But a holistic approach which allows an integrated modeling of requirements and systems during the mechatronic development process is currently missing. Especially the lack of formalization of modeled requirements and systems is one main drawback of current development approaches as it precludes the possibility for automated requirements and systems evaluations. For this reason, a novel modeling approach is outlined in this paper.

III. INTEGRATED REQUIREMENTS AND SYSTEMS MODELING TECHNIQUE

In the following, the integrated requirements and systems modeling technique, which was developed in the research project "IMoMeSA" [28], is presented. First, an overview of the modeling approach is given including an outlook to the intended quality assurance. On this basis, the individual contents of the modeling technique are explained in detail.

A. The Approach in a Nutshell

The key aspects of the model-based engineering approach are displayed in Figure 1. In the upper part the different phases of (mechatronic) systems engineering according to [29] are shown as well as the editor for mechatronic systems modeling, the "IMoMeSA Modeller", which is not focused further in this article, but represents an integral part of the approach. Then, in the center the contents of the modeling technique are illustrated, which serve the purpose of the respective phases. Note that the arrangement of the contents reflects the engineering workflow implied by our approach. Finally, at its bottom the figure shows the "IMoMeSA Analyzer", which is intended for quality assurance throughout mechatronic systems engineering. A first insight into the planned quality assurance mechanisms is given in [28].

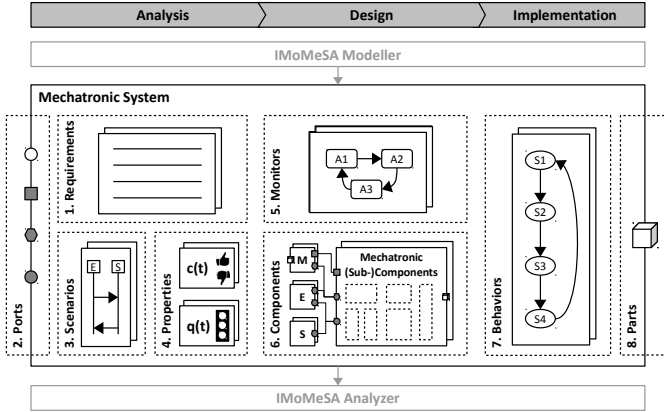


Fig. 1. Overview of the model-based approach including the engineering phases, the editor, the modeling contents and the quality assurance tool.

The analysis phase starts with modeling of textual and thus informal *requirements*, which define fundamental conditions of the mechatronic system. On this basis, the interface of the mechatronic system and its environment can be defined in terms of material, energy, data and generic flows. Therefore, respective input and output *ports* can be modeled in the proposed modeling technique.¹ Afterwards, these ports can be used to model so-called *scenarios*. Scenarios specify the required interaction between the system and its environment. Finally, *properties* can be added during analysis in order to formalize previously modeled requirements. Therefore, constraints and qualities can be applied. Both modeling elements use mathematical expressions (see Section III-B) to specify certain conditions the mechatronic system has to obey.

¹Typically, not all ports can be defined on the basis of modeled requirements. Some ports may also result from later modeling activities. Thus, the proposed engineering workflow defines only the general activity order, where iterations and jumps are allowed and even necessary.

While the analysis phase mainly deals with the modeling of the mechatronic system from the perspective of its environment, the design phase focuses on its inside. For this purpose, *monitors* can be specified defining the technical process to be implemented. In general, monitors describe required activities, which have to be performed by the mechatronic system. Subsequently, monitors need to be analyzed in order to decide whether the desired system behavior is too complex to be implemented directly. In this case new *components* can be created within the mechatronic system. Each component contains the same modeling contents as the mechatronic system itself. This creates a hierarchical structure of the mechatronic system and leads to a decomposition in detached and less complex modules. In addition to the previously described modeling contents, input and output ports of modeled components can be interconnected using channels.

In the implementation phase, behaviors and geometrical parts of the component or system under consideration can be modeled. Although *behaviors* as well as monitors rely on discrete-time I/O automata [30], both modeling elements can be distinguished by their usage: While monitors only observe input and output ports, behaviors set output ports in response to current and past input port values. Thereby, the actual system behavior can be specified and simulated. Furthermore, *parts* can be added to mechatronic or mechanic components. These parts define the geometrical representation of the component under consideration using simple rigid bodies.

B. The Modeling Contents in Detail

In the following, the individual contents of the proposed modeling technique are explained on meta-model level based on the meta-model specified in [31]. Especially the analysis phase is focused as this phase is most relevant for an integrated requirements and systems modeling. For the design and implementation phases brief descriptions and references to previously published work are provided instead.

1) *Analysis Phase*: As described in Section III-A, the analysis phase includes the declaration of the (mechatronic) component itself as well as the definition of requirements, ports, scenarios, and properties. Figure 2 provides an overview of the respective meta-model elements in UML notation [9] starting with the component on the top-most layer.

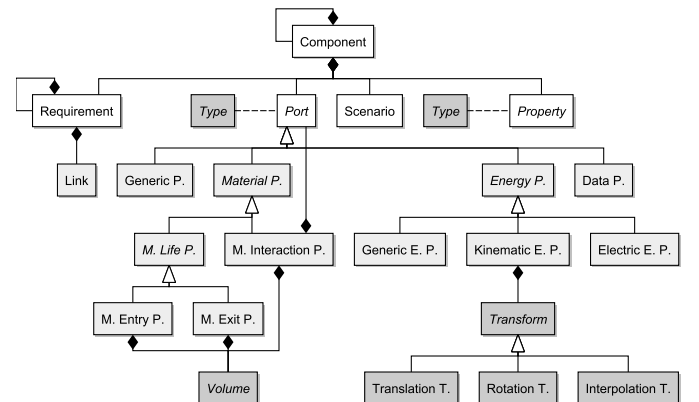


Fig. 2. Excerpt of the IMoMeSA meta-model including components, their direct children as well as the various port types.

The component represents the most central modeling element of the IMoMeSA approach. It can be used to declare the entire system under development as well as its mechatronic, mechanic, electric and software parts. For modeling customer demands, hierarchically structured requirements can be attached to components. At this point, no particular requirements description format is prescribed, but in the future requirements specification templates might be integrated [32]. To track the relationship between requirements and other model elements, links can be added to the model. Among others, links allow one to document design decisions (e.g. because of requirement X component Y was introduced), which is particularly useful in case requirements are changing and the system design has to be revised [33].

Ports are the first element used for requirements formalization. In contrast to other engineering approaches [34] four main port types are distinguished: Material, energy, data, and generic ports. Material ports are further divided into material life and interaction ports. Material life ports finally decompose into material entry and exit ports. Material entry and exit ports describe locations where material might appear or disappear during simulation. These ports are used for example in scenarios to describe material input and output requirements. In contrast, material interaction ports observe particular volumes for material presence during simulation. Furthermore, they allow one to specify an interface (in terms of ports) the observed component has to implement. Based on this interface dynamic interaction channels are maintained during simulation. Material interaction ports are used for example in monitors to constrain intermediate material states. On the other hand, energy ports are divided into kinematic, electric and generic energy ports. Kinematic energy ports represent the actuators of the mechatronic system and are defined using transforms (e.g. translation or rotation). Electric energy ports represent the interface of electric subsystem instead. Generic energy ports finally allow one to model any other kind of energy exchange (e.g. grinding or milling). In contrast, data ports are used to describe information flow within the software subsystem. Finally, generic ports allow one to model any kind of interaction not captured previously (e.g. operator inputs).

The remaining model contents for requirements formalization are properties and scenarios. Figure 3 provides a detailed look at both elements and their relationships.

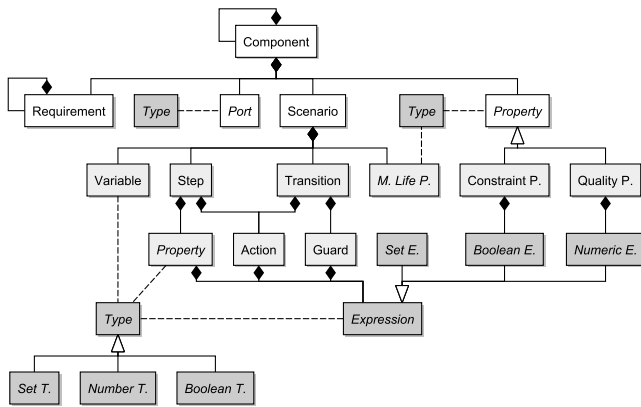


Fig. 3. Excerpt of the IMoMeSA meta-model including the types and expressions as well as scenarios and properties for requirement formalization.

Scenarios model the interaction of the environment with the mechatronic system. Consequently, scenarios provide inputs to the system under development, while processing its outputs. In general, the inputs and outputs are derived directly from the system ports. However, additional material life ports can be added to define scenario-specific material entry and exit locations (for an example see Section IV). Internally, scenarios are structured using steps and transitions. Steps model time intervals during which certain information is tracked and properties are measured. For example, a step might model the time interval after creating material at some material entry port and removing material at some material exit port. In contrast, transitions model the conditions that must hold for switching steps. For example, manipulated material could be required at some material interaction port to create new material at some material entry port. For information tracking within and across steps, scenarios allow one to specify variables including their respective type. Furthermore, within steps and transitions actions can be used to write variables (e.g. the cumulative energy consumption) and system inputs (e.g. some control input). Actions consist of type-compatible expressions, which might access system outputs as well as previous variable valuations. Finally, both on the component and on the scenario level properties can be used to express boolean constraints as well as numeric qualities over the system state based on port and variable valuations. For example, the maximum duration of scenario steps can be formulated as a constraint on some duration variable, while the energy efficiency can be expressed as a numeric quality over the cumulative energy consumption.

2) *Design Phase:* For the design phase the proposed approach provides two specific modeling elements: Monitors and subcomponents. Monitors allow one to define the technical process of the mechatronic system, while still leaving space for future design decisions. Subcomponents on the other hand can be used to refine the structure of the mechatronic system.

As described in [35], monitors are modeled in terms of observations, activities, transitions, guards, variables and actions as well as constraints and qualities. Observations provide streams of information about system execution, such as the location and state of workpieces or the energy consumption during some time interval. Practically, observations equal to the input and output ports of the mechatronic system. Then, activities, transitions and guards are used to structure the course of action of the technical process. Examples are waiting, milling and grinding activities during gear manufacturing as well as guards and transitions over intermediate workpiece state (obtained over observations). Subsequently, variables and actions are used to track information about system execution. For example, the duration of single activities or the energy consumption of activity sequences can be tracked. Constraints and qualities finally describe required and desired properties, which are reported to the developer during quality assurance. E.g. the maximum duration of activities can be constrained or the energy efficiency of activities can be measured.

Subcomponents on the other hand are modeled in terms of the same elements as the mechatronic system itself. Finally, channels can be used to specify the static interaction between the components. Note that during simulation also dynamic interactions might exist depending the material locations. Details about the underlying theory can be found in [36].

3) *Implementation Phase*: Finally, for the implementation phase mainly behaviors and parts are provided. Behaviors set the energy and data outputs based on respective material, energy and data inputs. Parts on the other hand model the geometric appearance of mechatronic or mechanic components.

For behavior modeling in principle a variety of formalism can be used such as input/output automata, table specifications and textual notations [37], [38]. Here, mainly input/output automata are used due to their expressiveness and familiar graphical notation. Similar to monitors, behaviors are described in terms of states, transitions and guards as well as variables and actions. States, transitions and guards can be used to structure the output behavior, while variables and actions allow to track information during behavior execution and trigger actual outputs. For example, states can be used to model active and defective states of sensors, while actions output active or defective data packets respectively. On the actuator side actions can be used to output kinematic energy along user-defined rotation or translation transforms. Note that this energy can be transferred to mechatronic or mechanic components either over predefined channels or over spontaneous bindings, which are explained in more detail in [36].

Finally, parts are described in terms of primitive volumes and their compositions. For example, primitive volumes can be cubes, spheres, cylinders and other basic shapes. Composition in this case means the union of several primitive and composite volumes. Note that the parts of some mechatronic or mechanic component are composed of the parts of all subcomponents.

IV. AN INDUSTRY-CLOSE CASE STUDY

In the following, the presented approach is evaluated using an industry-close case study. After a brief introduction of the application example and the evaluation method, the requirements and systems modeling is applied. Finally, the approach is evaluated with expert feedback from industry in order to ensure the suitability within further industrial applications.

A. Application Example and Evaluation Method

For benchmarking various mechatronic development methods, a miniaturized packaging machine is currently being set up at the Institute for Machine Tools and Industrial Management (*iwb*). The main function of this machine is to transform paper into stamped boxes, which are filled with desired goods. Within this machine, transport components are applied multiple times for the interconnection of individual packaging components. Their task, for example, is to bring paper blanks to some printing component, to align these blanks at a certain position and to finally dispatch the printed blanks.

To evaluate the presented approach, the integrated requirements and systems modeling technique is applied to such transport component (see Section IV-B). In addition, the experiences made in this study were discussed with experts from the machine tools industry. The aim of this discussion was to determine the *relevance of various aspects* for early mechatronic systems engineering phases and to evaluate the *aspect coverage* within the presented engineering approach. The result of this evaluation is presented in Section IV-C.

B. Mechatronic Requirements and Systems Modeling

In the following, the results of the modeling process regarding the mechatronic transport component are described. Therefore, the following explanations are subdivided into the analysis, design and implementation phase.

1) *Analysis of the Mechatronic Transport Component*: As proposed in Section III-A, the development of the transport component starts with the modeling of textual requirements. In the example the requirements include cycle times, energy consumption and material positions. The latter can be used directly for modeling the component's ports. Three different paper positions (i.e. *Start*, *Middle*, *End*) are defined using material interaction ports (see Figure 4). Additionally, one electric energy output port (i.e. *Power load*) and three data ports (i.e. *Feeded*, *Dispatch*, and *Dispatched*) for the communication with a top-level control are modeled. Note that no separate *Feed* data input port is needed, since the feeding process of a transport component has to be started autonomously when paper is inserted. Furthermore, an electric energy output port is used rather than an electric energy input port, because the energy demand of the transport component is modeled rather than the electric energy supply.

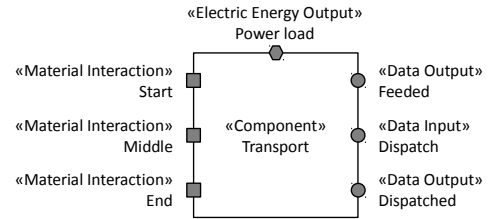


Fig. 4. Interface (i.e. ports) of the mechatronic transport component.

Then, based on the mechatronic interface a scenario is specified, which describes the interaction of the transport component with its environment (see Figure 5). Therefore, the scenario defines a material entry port, which is necessary to create paper during scenario execution. In the scenario, the environment first creates a paper at the respective material entry port. Consequently, the transport component should feed the paper to the *Middle* position and send the *Feeded* signal. Afterwards, the environment triggers the *Dispatch* signal, which finally should lead to paper being observed at the *End* location and the signal *Dispatched* being transmitted.

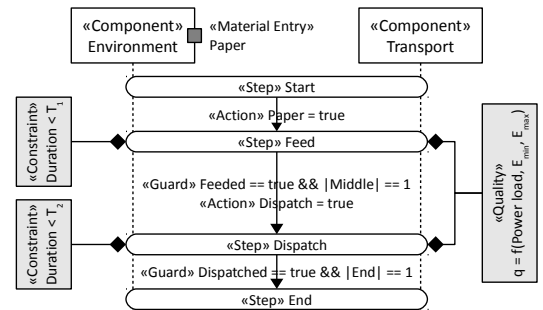


Fig. 5. Scenario for testing the mechatronic transport component.

In the next step, properties are added to selected steps of the scenario. In this example two constraints regarding the

cycle time and one quality regarding the energy consumption are modeled. The constraints formalize upper limits for the duration of feeding and dispatching, while the quality assesses the respective energy efficiency with respect to a theoretic minimum and maximum energy demand. Note that the minimum and maximum energy demand has been estimated by the responsible engineers, such that the meaning of the absolute quality values depends on the accuracy of this estimation.

2) Design of the Mechatronic Transport Component:

During design, a monitor for the technical process of the transport component is specified first (see Figure 6). The monitor consists of four activities and respective transitions. Monitoring starts in the *Wait* activity. If some material is detected at the *Start* position, the monitor switches to the *Feed* activity until material is observed at the *Middle* position and the *Feeded* signal has been sent. Then, the monitor switches to another *Wait* activity, until the *Dispatch* signal has been received. Subsequently, the *Dispatch* activity is executed, until the material is located at the *End* position and the *Dispatched* signal has been observed. Finally, the monitor switches back to the original *Wait* activity.

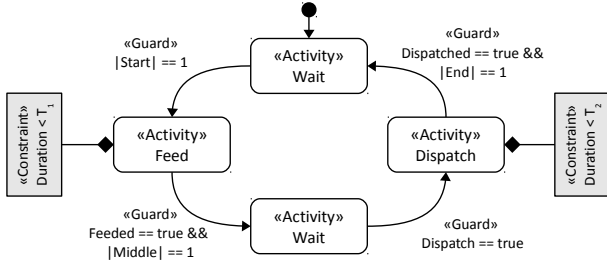


Fig. 6. Monitor of the mechatronic transport component.

In this example the monitor is considered too complex to be implemented directly by means of behaviors and parts. Thus, a decomposition into subcomponents is carried out. Figure 7 shows the resulting component structure consisting of light barriers at the *Start* and *End* position, a conveyor belt with an associated drive, a material stopper with an associated valve as well as a control with an associated timer. Due to space constraints, the compressor providing the necessary air pressure for the material stopper is omitted here. Also the *Power load* energy output port is not discussed further.

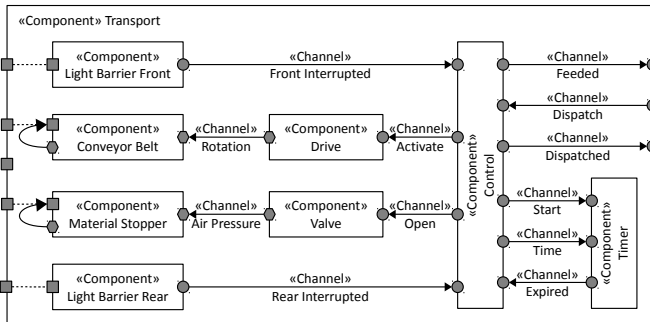


Fig. 7. Decomposition of the mechatronic transport component.

Subsequently, the analysis and design steps need to be carried out for the subcomponents in order to determine

whether further decomposition is necessary. In this example it is sufficient to point out that all components shown in Figure 7 turned out to be atomic, which means that their implementation can be modeled without further decomposition.

3) *Implementation of the Mechatronic Transport Component:* As mentioned in Section III, the implementation of components can be carried out using behaviors and parts. Here, the behaviors of the control and valve components as well as the parts of the conveyor belt component are presented.

Since modeling of parts is limited intentionally to base geometries, the conveyor belt is represented by a simple cuboid defining its approximate dimensions (see 8a). In addition, a translation defining the motion direction of material colliding with the conveyor belt is modeled and displayed. In contrast, the behavior of the conveyor belt is defined using an I/O automaton generating kinematic energy (not shown). The kinematic energy is transferred to colliding paper using a material interaction port on top of the conveyor belt cuboid (see the box in Figure 8a and [36] for more information). In Figure 8b the behavior of the valve is displayed instead. A single state *Active* and two transitions are sufficient to describe its behavior: If the input signal *Open* is observed, a constant is applied to the energy output port *Air Pressure*. If the input signal *Open* is not observed, the energy output port *Air Pressure* is set to zero instead. Finally, Figure 8c shows the discrete-time I/O automaton of the control component. Note that the structure of the control behavior is equivalent to the structure of the monitor from the previous section. One difference, for example, is the use of data inputs from specific sensors instead of using observations of material interaction ports.

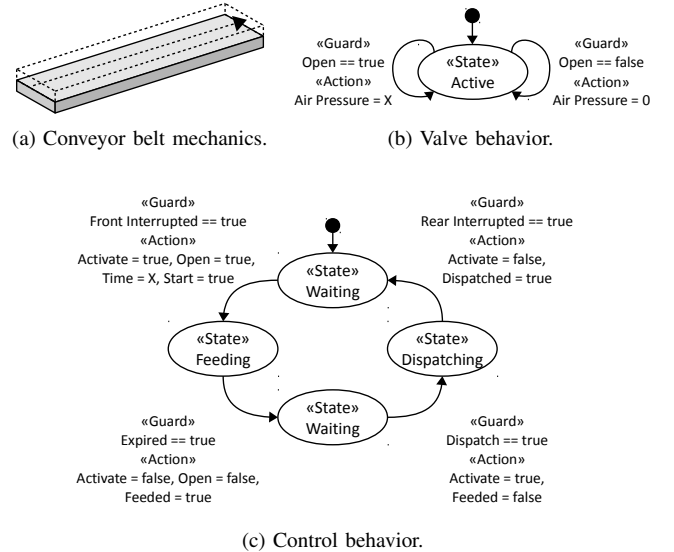


Fig. 8. Implementation of the mechatronic transport component.

Similar implementations have been developed for the light barriers, the drive, the material stopper and the timer. Overall, the implementations form a completely executable mechatronic system model, which can be used for syntactic and semantic quality assurance as discussed in [28].

C. Evaluation

As mentioned before, the purpose of this evaluation was to answer the question whether aspects, which are generally relevant for mechatronic systems engineering, are covered by the presented modeling approach. Therefore, the modeling technique as well as its application to the miniaturized packaging machine were discussed in several workshops with a total of nine experts from various companies of the machine tools industry. The result of this discussion are summarized in Table I, which compares the *relevance* (divided in "strong" and "weak") with the *coverage* (divided in "yes" and "no") of engineering aspects for mechatronic systems².

TABLE I. COMPARING RELEVANCE AND COVERAGE OF DIFFERENT ASPECTS DURING EARLY PHASES OF MECHATRONIC SYSTEMS ENG.

		Aspect coverage	
		yes	no
Aspect relevance	strong	- Acceptance tests - Customer requirements - Modular system structures - Processes and behaviors	- Informal schematic diagrams - Mechatronic construction kits - Variant / change management - Failure mode modeling
	weak	- Completely formalized models - Geometry modeling	- Discipline-specific construction - Continuous-time behavior - Physical effects - Hydraulics / pneumatics

The upper left quadrant of Table I lists aspects, which are considered to be strongly relevant and are covered by the presented modeling technique. One central aspect of early development phases is the coordination with customers. In particular, acceptance tests as well as fundamental requirements of the system to be developed (e.g. concerning cycle times) have to be specified. Therefore, the modeling technique allows the creation of scenarios for the entire system or its components as well as the modeling of requirements or properties. In addition, the general structure of the mechatronic system as well as its basic processes and functions have to be defined. Especially the modular set-up of mechatronic systems is currently gaining importance as a result of growing mass customization. For this purpose, mechatronic or discipline-specific components with their associated ports can be used to encapsulate the system's functionality in less complex and detached modules. Furthermore, monitors can be added to modeled components in order to specify (technical) processes, while behaviors are used to define the actual implementation of a certain component.

On the other hand, the case study also showed that several aspects covered by the modeling technique are not completely purposeful for early phases of mechatronic systems engineering (lower left quadrant of Table I). In particular, the complete formalization of models (necessary for semantic correctness checks) takes a lot modeling effort, which is typically not available due to prevailing time and cost pressures. In addition, the proposed geometry modeling is too complex for early engineering phases, in which rather easy-to-draw geometries known from "Mechatronic Sketching" [40] are required.

Furthermore, also aspects could be identified, which are relevant for early mechatronic systems engineering, but are not

covered by the modeling technique yet (upper right quadrant of Table I). Among these, the drawing of informal schematic diagrams is listed first. In early development phases simple drawings and diagrams are typically used to explain various aspects of the mechatronic system among participating disciplines (e.g. principle sketch of a conveyor belt's function). Furthermore, the application of a mechatronic construction kit can improve the overall engineering approach, since the reuse of components (e.g. via templates) is essential to lower development times and costs. Another important aspect is the integration of variant and change management aspects within the modeling technique. Since the trend of mass customization is currently emerging, the creation of variants and the immediate reaction to changing customer demands becomes increasingly important. Thus, the presented modeling technique should provide mechanisms, which simplify these tasks, in future. Another drawback of the current modeling technique is the aspect that only the normal performance of mechatronic systems can be modeled. In order to reduce malfunctions during system usage, possible system failures should be considered as soon as possible. Therefore, mechanisms have to be invented to model and analyze failures parallel to the actual system development.

Finally, also aspects are listed in the lower right quadrant of Table I, which were intentionally not integrated within the presented modeling technique as these aspects are typically not relevant until later phases of the development process. In this discussion the discipline-specific construction (e.g. the detailed mechanical CAD), the modeling of continuous-time behavior, the depiction of physical effects (e.g. friction) as well as the representation of hydraulics and pneumatics were identified.

V. SUMMARY & OUTLOOK

In this paper a model-based approach for an integrated requirements and systems modeling in the mechatronic development process was introduced. After an overview of the engineering approach, the individual modeling contents were presented on a meta-model level. Subsequently, the technique was applied to an industrial-close case study to show the feasibility. Through an evaluation of this example with experts from the machine tools industry the strengths and weaknesses of the presented modeling technique could be identified.

Pending research activities address the outlined drawbacks: The modeling technique is currently revised in terms of a better scalability in order to allow a simpler and less formal modeling. Furthermore, the modeling technique will be enhanced by the relevant aspects for early mechatronic systems engineering, which are currently not covered. Thereby, it is a major challenge to embed new modeling concepts appropriately to keep the number of applied modeling concepts to a minimum and, thus, preserve an intuitive modeling of mechatronic systems. In addition, the implementation of quality assurance mechanisms will be one key factor to justify the high modeling effort. Finally, further research tasks will address the application of the entire approach to an industrial case study (e.g. a machine tool). In contrast to the shown example, the complexity will be significantly larger, which influences the evaluation especially regarding the necessary modeling effort. In particular, it must be evaluated whether the modeling will be as intuitive and helpful as it was demonstrated for the transport component.

²These engineering aspects were identified in accordance to the Mechatronic Reference Model presented in [39].

ACKNOWLEDGEMENTS

The authors would like to thank the German Federal Ministry of Economics and Technology (BMWi) for the sponsoring of the research project IMoMeSA (Funding code: 435 ZN).

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